SDS 2412: Analysis of Large Datasets

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1. Overview

- This is a milestone-driven program in which each student group selects one large-scale data problem and develops it into a complete, end-to-end distributed data and machine learning system.
- The system evolves through six structured milestones, progressively extending the same pipeline into a scalable and research-oriented data system.
- The core objectives include mastery of:
  - Distributed data systems
  - Large-scale data processing
  - Scalable machine learning
  - Data engineering and pipelines
  - AI system deployment and optimization

2. Topic Selection (Mandatory)

- Each group must select one system from the following:
  - Real-Time Financial Fraud Detection System
  - Large-Scale Recommendation System
  - Social Media Trend Analysis System
  - Climate Data Analysis and Prediction System
  - Healthcare Analytics and Disease Prediction System
  - Smart City Data Intelligence System
  - Cybersecurity Threat Detection System
  - Large-Scale Log Analytics System
  - Supply Chain Optimization System
  - Energy Consumption Prediction System
- **Rules:**
  - The topic cannot be changed after selection.
  - All milestones must evolve the same system.
  - Fragmented or separate systems are not allowed.

3. Core Structure: Six Evolutionary Milestones

Milestone 1: Data Foundations & System Architecture (Weeks 1–3)

Focus: Translating real-world problems into large-scale data systems.

Requirements:

- Problem definition and data characterization (volume, velocity, variety)
- Data sourcing and ingestion strategy
- Complexity and scalability analysis (Big O concepts)
- System architecture design (Lambda or Kappa)
- Initial batch data pipeline

Deliverables:

- Data ingestion prototype
- System architecture design
- Data specification and sourcing report

Milestone 2: Distributed Data Processing (Weeks 4–6)

Focus: Building scalable processing systems.

Requirements:

- Data partitioning and sharding
- Batch processing (MapReduce or Spark concepts)
- Distributed storage (HDFS or equivalent)
- Fault tolerance and replication
- Job scheduling and execution pipelines

Deliverables:

- Distributed processing pipeline
- Workflow documentation
- Scalability and performance analysis

Milestone 3: Streaming & Real-Time Systems (Weeks 4–6 Extended)

Focus: Handling high-velocity data streams.

Requirements:

- Streaming ingestion (Kafka or equivalent)
- Event-driven architecture
- Real-time analytics pipeline
- Approximate algorithms (Bloom filters, Count-Min Sketch)
- Batch vs streaming comparison

Deliverables:

- Real-time processing system
- Latency and throughput evaluation
- Streaming system documentation

Milestone 4: Scalable Machine Learning & Analytics (Weeks 7–9)

Focus: Machine learning at scale.

Requirements:

- Feature engineering on large datasets
- Model training (e.g., boosting, neural networks)
- Distributed or parallel training
- Model validation and evaluation
- Basic explainability

Deliverables:

- Trained model
- Evaluation metrics
- Performance report

Milestone 5: System Optimization & Deployment (Weeks 10–12)

Focus: Operationalization and optimization.

Requirements:

- Model deployment pipeline
- Monitoring and drift detection
- Pipeline optimization (caching, partitioning)
- Integration with NoSQL systems
- Workflow orchestration

Deliverables:

- Deployed system
- Monitoring and logging framework
- Optimization report

Milestone 6: Integrated Intelligent System & Capstone (Weeks 13–14)

Focus: Full system integration and innovation.

Requirements:

- Integration of batch, streaming, and ML components
- End-to-end pipeline (data to output)
- At least one system-level innovation
- Analytical outputs and interpretation
- System evaluation

Deliverables:

- Fully functional large-scale system
- Final technical report
- Research-oriented enhancement
- Live demonstration

4. Evaluation Structure

- Block 1: Data modeling and architecture
- Block 2: Distributed processing
- Block 3: Streaming computation
- Block 4: Scalable machine learning
- Block 5: Deployment and optimization
- Block 6: Integration and research contribution

5. Non-Negotiable Rules

- One topic per group
- One continuous evolving system
- No restarting previous work
- No black-box use of frameworks
- Each milestone must document:
  - What changed
  - What failed
  - Why it failed
  - How it was fixed

6. Final Output Requirements

- Fully working large-scale data system
- Complete and documented pipelines
- Technical report including:
  - Problem definition
  - System evolution
  - Architecture and implementation
  - Performance and scalability evaluation
- Final live demonstration and oral defense

Conclusion

- Work within existing groups.
- Weekly progress presentations are mandatory.